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Alessandro Regorda

Date of birth: 21/10/1982 | **Nationality:** Italian | **Gender:** Male | **Phone number:** (+39) 3285634106 (Mobile) | **Email address:** alessandro.regorda@protonmail.ch | **Email address:** alessandro.regorda@unimi.it | **Website:** https://github.com/aleregorda | **Website:** https://orcid.org/0000-0002-7677-4542 | **Website:** https://aleregorda.github.io/

EDUCATION AND TRAINING

01/2014 – 12/2016

PH.D. Università degli Studi di Milano (Italy) - Université de Nice-Sophia Antipolis (France)

Ph.D. in co-tutorship between Università degli Studi di Milano and Université de Nice-Sophia Antipolis (tutors Prof. Anna Maria Marotta, Prof. Maria Iole Spalla and Prof. Jean-Marc Lardeaux), focused on:

- the implementation of Fortran code *SubMar* verifying the impact of various mechanisms on the thermo-dynamics of the subduction zones;
- the analysis of the distribution in time and space of P-T and metamorphic facies predicted by the models;
- the comparison with natural data of the Variscan metamorphism preserved in the Alps and in the French Central Massif.

Field of study Earth Sciences | **Level in EQF** EQF level 8 |

Thesis The thermo-mechanical evolution of the subduction-collision systems

10/2010 – 04/2013

M.SC. IN EARTH SCIENCES Università degli Studi di Milano

- Thesis: The original Fortran code *SubMar* has been implemented with the introduction of new mechanisms taking place in the subduction systems during an ocean-continent active convergence (supervisors Prof. Anna Maria Marotta and Prof. Maria Iole Spalla).
- Internship: Georeferencing and digitalising of geological maps using *ArcGis*.
- Courses: English, Seismology, Fluid dynamics, Physics of the solid Earth, Geodynamics, Numerical modelling of geodynamics processes, Structural analysis, Microstructures of crystalline rocks.

Final grade 110/110 cum laude | **Level in EQF** EQF level 7 |

Thesis Numerical model of oceanic and continental subduction and comparison with natural data of the Variscan orogenesis

09/2007 – 10/2010

B.SC. IN GEOLOGY Università degli Studi di Milano

- Thesis: A garnet pyroxene amphibolite sample has been studied by means of micro and mesostructural analysis, with focus on the relationship between mineral growth and deformation. A 3D model of the sample has been built using software *Move2010.1* (supervisor Prof. Michele Zucali).
- Internship: Geoelectrical exploration (SEV and ERGI) of the hydrostratigraphy of an area in the Po plain (Northern Italy).
- Courses: Mathematics, Physics, Chemistry, Geochemistry, Geophysics exploration, Physics of the Earth, Petrology, Mineralogy. Field school (Dent Blanche nappe, Western Alps, Italy) focus on geological and structural maps.

Final grade 108/110 | **Level in EQF** EQF level 6 |

Thesis Analysis and 3D modelling of a garnet pyroxene amphibolite sample of the Valpelline series (Western Alps, Italy)

WORK EXPERIENCE

TENURE TRACK RESEARCHER – UNIVERSITÀ DEGLI STUDI DI MILANO – 01/05/2024 – Current – MILAN, ITALY

- maintenance of computers

- relations with suppliers
- coaching a junior Ice Hockey team (10 hours/week)

POST-DOC IN EARTH SCIENCE – UNIVERSITÀ DEGLI STUDI DI MILANO – 01/01/2024 – 30/04/2024 – MILAN, ITALY

'Modelling of gravitational effects from tectonics processes of different time scales' as part of the NGGM-MAGIC project (PI Prof. Anna Maria Marotta), focusing on the analysis of gravitational anomalies predicted by 2D and 3D models in response to tectonics deformation, and their comparison with real data.

SCIENCE AND MATHEMATICS TEACHER LOWER SECONDARY SCHOOL – STATE SCHOOL 'ALDA MERINI' – 03/10/2022 – 31/12/2023 – MILAN, ITALY

INDEPENDENT CONTRACTOR – UNIVERSITÀ DEGLI STUDI DI MILANO – 04/08/2022 – 23/08/2022 – MILAN

Worked as an independent contractor at the Department of Earth Science 'Ardito Desio' for the construction of a database of rheological parameters of different rocks and for the calibration of an algorithm for the implementation of a non-linear rheology in the numerical code *SubMar* (PI Prof. Roberto Sabadini and Prof. Anna Maria Marotta).

POST-DOC IN EARTH SCIENCE – UNIVERSITÀ DEGLI STUDI DI MILANO – 01/07/2018 – 30/06/2022 – MILAN, ITALY

'Numerical modelling of subduction systems and analysis of the corresponding gravimetric signatures' (supervisor Prof. Anna Maria Marotta), focusing on:

- Analysis of metamorphic facies and deformation fabrics within subduction complexes;
- Comparison between prediction of the models in terms of signatures on the gravity field and P-T(-t) conditions with natural data from recent data of gravity anomalies and P-T estimates of Variscan metamorphism;
- Teaching assistance of the course 'Numerical modelling of geodynamic processes'.

During the Post-Doc I developed a brand new 2D FEM code coined *FALCON* (Finite element ALgorithm for COmputational aNalysis) characterised by a non-linear visco-plastic behaviour. The code has been thoroughly benchmarked with community benchmarks and analytical solutions:

- benchmarks performed at https://github.com/aleregorda/Benchmarks;
- code description at https://github.com/aleregorda/Code_description/blob/main/main.pdf.

SCIENCE AND MATHEMATICS TEACHER LOWER SECONDARY SCHOOL – STATE SCHOOL 'MANZONI' – 27/09/2017 – 30/06/2018 – PARABIAGO, ITALY

INDEPENDENT CONTRACTOR – UNIVERSITÀ DEGLI STUDI DI MILANO – 27/09/2013 – 27/10/2013 – MILAN, ITALY

Worked as an independent contractor at the Department of Earth Science 'Ardito Desio' for the construction of a database of natural P-T data of Variscan metamorphism in the Alps and comparison with data predicted by a thermo-mechanical model (PI Prof. Roberto Sabadini and Prof. Anna Maria Marotta).

LANGUAGE SKILLS

Mother tongue(s): **ITALIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	B2	C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Operating System

Proficient user of Linux (Debian, Ubuntu) and Windows | Independent user of Linux (Fedora)

Programming languages

Proficient user of Fortran | Basic user of Matlab/GNU Octave

Software

Independent user of Microsoft Office package, LaTeX Editor, ParaView, Gnuplot and GMT | Basic user of ArcGis/QGis

PUBLICATIONS

2026
Alessandro Regorda, Marco Filippi, Manuel Roda, Fabiola Caso, Federico Farina, Chiara Benedetta Piloni, Michele Zucali (2026). Post-Collisional Evolution in the Southern Variscan Belt: Insights From Numerical Modeling of the Valpelline Series (Western Alps). Tectonics, 45(8), e2026TC009377, doi: 10.1029/2026TC009377

2026
Lorenzo Rossi, Alessandro Regorda, Mirko Reguzzoni, Carlo Iapige De Gaetani, Riccardo Barzaghi, Roberto Sabadini, Anna Maria Marotta (2026). Sensitivity of the MAGIC satellite instrumentation to the gravity rate of change due to rifting in the Gulf of Aden. Geophysical Journal International, 247(1), ggag237, doi: 10.1093/gji/ggag237

2025
Anna Maria Marotta, Valeria Fedeli, Alessandro Regorda, Roberto Sabadini (2025). Gravity rate of change due to slow tectonics: insights from numerical modeling. Geosciences, 15(9). doi: 10.3390/geosciences15090359

2024
Alessandro Regorda, Manuel Roda (2024). Thermo-mechanical effects of microcontinent collision on ocean-continent subduction system. Journal of Geophysical research: Solid Earth, 129(12). doi: 10.1029/2024jb029908

2023
Anna Maria Marotta, Riccardo Barzaghi, Arcangela Bollino, Alessandro Regorda, Roberto Sabadini (2023). The gravitational signature of the dynamics of oceanization in the Gulf of Aden. Tectonophysics, 869, 230110, doi: 10.1016/j.tecto.2023.230110

2023
Manuel Roda, Maria Iole Spalla, Marco Filippi, Jean-Marc Lardeaux, Gisella Rebay, Alessandro Regorda, Davide Zanoni, Michele Zucali, Guido Gosso (2023). Metamorphic Remnants of the Variscan Orogeny across the Alps and Their Tectonic Significance. Geosciences 13, 300, doi:10.3390/geosciences13100300

2023
Davide Zanoni, Marco Filippi, Manuel Roda, Alessandro Regorda, Maria Iole Spalla (2023). Alpine convergence record in the Carboniferous Badstub Formation, Upper Austroalpine basement nappes, Austria. International Geology Review, doi:10.1080/00206814.2023.2206443

2023
Alessandro Regorda, Cedric Thieulot, Iris van Zelst, Zoltan Erdos, Julia Maia, Susanne Buiter (2023). Rifting Venus: Insights from Numerical Modeling. JGR Planets, 128(3), doi: 10.1029/2022JE007588

2022
Marco Filippi, Davide Zanoni, Gisella Rebay, Manuel Roda, Alessandro Regorda, Jean-Marc Lardeaux Maria Iole Spalla (2022). Quantification of Alpine Metamorphism in the Edolo Diabase, Central Southern Alps. Geosciences, 12(8), 312, doi:10.3390/geosciences12080312

2022
Arcangela Bollino, Alessandro Regorda, Roberto Sabadini, Anna Maria Marotta (2022). From rifting to oceanization in the Gulf of Aden: Insights from 2D numerical models. Tectonophysics, 838, 229483, doi:10.1016/j.tecto.2022.229483

2021
Alessandro Regorda, Maria Iole Spalla, Manuel Roda, Jean-Marc Lardeaux, Anna Maria Marotta (2021). Metamorphic Facies and Deformation Fabrics Diagnostic of Subduction: Insights From 2D Numerical Models. Geochemistry, Geophysics, Geosystems, 2021, 22(10), doi:10.1029/2021GC009899

2020
Anna Maria Marotta, Federica Restelli, Arcangela Bollino, Alessandro Regorda, Roberto Sabadini (2020). The static and time-dependent signature of ocean-continent and ocean-ocean subduction: the case studies of Sumatra and Mariana complexes. Geophysical Journal International, 221(2), 788–825, doi:10.1093/gji/ggaa029

2020
Manuel Roda, Michele Zucali, Alessandro Regorda, Iole Spalla (2020). Formation and evolution of a subduction-related mélange: The example of the Rocca Canavese Thrust Sheets (Western Alps). GSA Bulletin, doi:10.1130/B35213.1

2020
Alessandro Regorda, Jean-Marc Lardeaux, Manuel Roda, Anna Maria Marotta, Maria Iole Spalla (2020). How many subductions in the Variscan orogeny? Insights from numerical models. Geoscience Frontiers, doi:10.1016/j.gsf.2019.10.005

2019
Manuel Roda, Alessandro Regorda, Maria Iole Spalla, Anna Maria Marotta (2019). What drives Alpine Tethys opening: clues from the review of geological data and model predictions. Geological Journal, doi:10.1002/gj.3316

2017
Alessandro Regorda, Manuel Roda, Anna Maria Marotta, Maria Iole Spalla (2017). 2-D numerical study of hydrated wedge dynamics from subduction to post-collisional phases. Geophysical Journal International, 211, 974-1000, doi:10.1093/gji/ggx336

2013
Alessandro Regorda, Anna Maria Marotta, Maria Iole Spalla (2013). Numerical model of an ocean/continent subduction and comparison with Variscan orogeny natural data. Rendiconti Online Societa Geologica Italiana, 29,142-145

CONFERENCES AND SEMINARS

15/09/2026 – 17/09/2026 Padova
SGI-SIMP Annual congress

- Oral presentation, “Linking petrostructural relics and subduction dynamics through numerical modelling” - Regorda, A., Roda, M. & Spalla, M.I.
- Poster, “Investigating the evolution of Venusian rifts through integrated observations and modelling” - Regorda, A., van Zelst, I., Maia, J., Erdős, Z. & Thieulot, C.

21/06/2026 – 26/06/2026 Seč
Ada Lovelace Workshop

Poster "Understanding the Evolution of Venusian Rifts through Integrated Observations and Modeling” - Regorda, A., van Zelst, I., Maia, J., Erdős, Z. & Thieulot, C.

03/05/2026 – 08/05/2026 Wien

EGU general assembly:

- Poster, “2D numerical models of the Variscan post-collisional evolution: example from the Valpelline Series (western Alps)” - Regorda, A., Filippi, M., Roda, M., Caso, F., Piloni, C.B., Farina, F. & Zucali, M.
- Poster, “Constraining the evolution of Venusian rifts: an integrated observation and modelling approach” - Thieulot, C., Regorda, A., van Zelst, I., Maia, J. & Erdős, Z.

10/02/2026 – 13/02/2026 Udine

44th GNGTS National Conference

Oral presentation, “Numerical models of the post-collisional evolution of the southern Variscan belt: comparison with the Valpelline Series (western Alps)” - Regorda, A., Filippi, M. & Roda, M.

16/09/2025 – 18/09/2025 Padova

SGI-SIMP Annual congress

Oral presentation, “Post-collisional evolution in extensional tectonic regime: Insights from numerical models” - Regorda, A., Filippi, M. & Roda, M.

07/09/2025 – 12/09/2025 Lisbon

GeoMod

Oral presentation, “2D numerical model of post-collisional tectonic evolution: example from the Permian Valpelline Series, western Alps” - Regorda, A., Filippi, M. & Roda, M.

27/04/2025 – 02/05/2025 Wien

EGU general assembly

Poster, “2D numerical analysis on microcontinents subductibility: subduction or collision” - Regorda, A. & Roda, M.

11/02/2025 – 14/02/2025 Bologna

43rd GNGTS National Conference

Oral presentation, “Thermo-mechanical effects of microcontinent subduction” - Regorda, A. & Roda, M.

01/09/2024 – 06/09/2024 Montpellier

Ada Lovelace Workshop

Poster, “Thermal effects of microcontinent subduction: insight from 2D numerical models” - Regorda, A. & Roda, M.

23/06/2024 – 25/06/2024 Chieti

CRUST Workshop

Poster, “2D numerical simulations of micro-continents collision in ocean-continent subduction systems” - Regorda, A. & Roda, M.

26/09/2023 – 28/09/2023 Paris

GeoMod

Poster: "2D numerical simulations of micro-continents collision in ocean-continent subduction systems" - Regorda A. & Roda M.

05/04/2023 Online

NASA Venus Seminar

Invited oral presentation "The Rifting process on Venus: Insights from Numerical Modeling" - Regorda A., Thieulot C., van Zelst I., Erdos Z., Maia J. & Buiter S.

27/06/2022 – 29/06/2022 Trieste

GNGTS Annual congress

Poster, "Parametric study of the effects of micro-continents collision in a oceanic subduction systems by 2D numerical simulations" - Regorda A., Roda M. & Thieulot C.

01/11/2021 – 05/11/2021 Lyon

Réunion des Sciences de la Terre Annual congress

Oral presentation, "How many subductions in the Variscan orogeny? Insights from numerical models" - Regorda A., Lardeaux J.-M., Roda M., Marotta A.M. & Spalla M.I.

14/09/2021 – 16/09/2021 Online

SGI Annual congress

Oral presentation, "Metamorphic P-T conditions characteristic of subduction/collision systems: insights from 2D numerical models" - Regorda A., Spalla M.I., Roda M., Lardeaux J.-M. & Marotta A.M.

22/06/2021 – 24/06/2021 Online

GNGTS Annual congress

Poster, "Metamorphic and deformation patterns produced during an oceanic subduction: insights from 2D numerical models" - Regorda A., Spalla M.I., Roda M., Lardeaux J.-M. & Marotta A.M.

12/11/2019 – 14/11/2019 Rome

GNGTS Annual congress

Oral presentation, "How many subductions in the Variscan orogeny? Insights from numerical models" - Regorda A., Lardeaux J.-M., Roda M., Marotta A.M. & Spalla M.I.

16/09/2019 – 19/09/2019 Parma

SGI-SIMP-SOGEI Annual congress

Poster, "How many subductions in the Variscan orogeny? Insights from numerical models" - Regorda A., Lardeaux J.-M., Roda M., Marotta A.M. & Spalla M.I.

14/12/2015 – 18/12/2015 San Francisco

AGU Fall Meeting

Oral presentation, "New insights into the dynamics of wedge areas from a 2D numerical study of the effects of shear heating and mantle hydration on an ocean-continent subduction system" - Roda M., Regorda A., Marotta A.M. & Spalla M.I.

30/09/2015 Catania

GIGS Annual meeting

Oral presentation, "2D numerical model of an ocean/continent subduction system: examples from the Variscan crust" - Regorda A., Marotta A.M., Roda M., Lardeaux J.M. & Spalla M.I.

09/06/2015 – 11/06/2015 Rennes

Variscan 2015

Oral presentation, "Effects of mantle hydration and viscous heating on the dynamics of mantle wedge in a subduction system: differences and similarities of 2D model predictions with examples from the Variscan crust" - Regorda A., Marotta A.M., Roda M., Lardeaux J.M. & Spalla M.I.

10/09/2014 – 12/09/2014 Milan

SGI-SIMP Annual congress

Oral presentation, "2D numerical study of the effects of mantle hydration and viscous heating on the dynamics of the wedge area within an ocean/continent subduction complex: the case study of Variscan crust in the Alpine domain" - Regorda A., Marotta A.M., Spalla M.I., Roda M. & Rebay G.

28/04/2014 – 02/05/2014 Wien

EGU General assembly

Poster, "Numerical modelling of an ocean/continent subduction and comparison with Variscan orogeny real data" - Marotta A.M., Regorda A., Spalla M.I., Roda M. & Rebay G.

28/10/2013 – 29/10/2013 Milan

GIGS Annual meeting

Oral presentation "Numerical modelling of an ocean/continent subduction and comparison with Variscan orogeny real data" - Regorda A., Marotta A.M. & Spalla M.I.

HOBBIES AND INTERESTS

Since 1998 basketball player for amateur associations. Enjoy mountaineering, hiking, climbing, skiing, camping.

TEACHING AND TUTORING ACTIVITIES

01/05/2024 – 30/09/2026
Co-tutor of Valeria Fedeli, Ph.D. student, Università degli Studi di Milano - 'Unraveling subduction zone initiation: geodynamic numerical modeling and gravity signature analysis'

01/10/2019 – 30/09/2022
Co-tutor of Arcangela Bollino, Ph.D. student, Università degli Studi di Milano - 'Quasi static gravity signatures in slow tectonic zones: assimilation of novel aerospace data and geophysical modeling'

2019
Co-supervisor of Federica Restelli, M.Sc. student, Università degli Studi di Milano - 'Effetti statici e dinamici della subduzione sul campo gravitazionale terrestre'

02/2025 – CURRENT
Course Coordinator of the M.Sc. course 'Numerical modelling of geodynamic processes' - Università degli Studi di Milano.

10/2019 – 12/2019
Teaching assistance of the M.Sc. course 'Numerical modelling of geodynamic processes' (Prof. Anna Maria Marotta) - Università degli Studi di Milano.

DEPARTMENTAL DUTIES

11/2025 – CURRENT
Researchers' Representative on the Steering Committee of the Faculty of Science and Technology at the Università degli Studi di Milano

11/2025 – CURRENT
Researchers' Representative on the Departmental Board at the Department of Earth Science 'Ardito Desio', Università degli Studi di Milano

2020 – 06/2022
Representative for the Department of Earth Science and Board member in the Council of Research Fellows of the Università degli Studi di Milano

2018 – 06/2022
Research fellows representative at the Department of Earth Science 'Ardito Desio', Università degli Studi di Milano

2014 – 2016
PhD students representative at the Department of Earth Science 'Ardito Desio', Università degli Studi di Milano

REVIEWING ACTIVITIES

Reviewer of Earth and Planetary Science Letters

Reviewer of Frontiers in Earth Science

Reviewer of Scientific Reports - Nature

Reviewer of Geophysical Journal International

PARTICIPATION IN FUNDED PROJECTS

01/01/2024 – 30/06/2026

Unit member of the NGGM-MAGIC Research Unit (Principal Investigator Prof. Anna Maria Marotta)

2018 – 2019

Unit member in 'Gravitational Seismology', ESA Endorsement (Principal Investigator Prof. Roberto Sabadini)

2013 – 2016

Unit member in MIUR-PRIN 2011 project ‘Birth and death of oceanic basins: geodynamic processes from rifting to continental collision in Mediterranean and Circum-Mediterranean orogens’ (Principal Investigator Prof. Maria Iole Spalla)

2007 – 2012

Unit member in 'SISMA' (Seismic Information System for Monitoring and Alert), funded by the Italian Space Agency (Principal Investigator Prof. Roberto Sabadini)